ARUSHA REGION - F2 MOCK EXAMINATION MAY 2024

B. MATHS - MARKING SCHEME.

1. (a) Prime numbers btn 70-90

71, 73, 83, 89 & 79

Sum = 71+73+79+83+89 - - - - 02

= 395 - - - - - 01

(b) $5/7$ → boys

Girls = 120

boys + Girls = $7/7$

$5/7$ + Girls = $7/7$

Girls = $7/7 - 5/7 = 2/7$ - - - - 01

Let the total number of students be x

$\frac{2}{7} \times x = 120$

$x = \frac{120 \times 7}{2}$

$x = 420$ - - - - 02

Total number of students = 420.

(c) 5%, $1/4$, 0.1, $3/5$

5% → 5%

$\frac{1}{4} \times \frac{25}{100} \rightarrow 25\%$

$0.1 \times 100 = 10\%$

$\frac{3}{5} \times \frac{25}{100} = 75\%$

02

In descending order = $3/5$, $\frac{1}{4}$, 0.1, 5%

02

2 (a) 06:30

(i) 12-hour clock = 06:30 PM. ... 01½

(ii) 24-hour clock = 1830 hours. ... 01½

(b) 1 kg = 1000 g

x = 4800 g

$x = \frac{4800}{1000}$

$= 4.8$

4800 g = 4.8 kg. ... 02

(ii) 1 tonne = 1000 kg

0.05t = x

x = 50 kg

0.05t = 50 kg. ... 02

(c) 350 ml × 3 = 1050 ml ... 01

1 litre = 1000 ml ... 01

Anna drink more juice.

Difference = 1050 − 1000 = 50 ml. ... 01

3. (a) $B\hat{A}C = A\hat{C}E = 50° = x$ (Alternate int. ∠s) AB // CE

$\therefore x = 50°$. 02

$A\hat{B}C = E\hat{C}D = 70° = z°$ (corresponding ∠s)

$z = 70°$ 02

$x + y + z = 180°$ (straight angle)

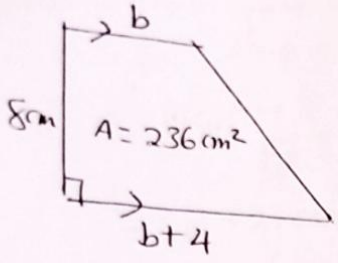
$50 + y + 70 = 180$

$y + 120 = 180$

$y = 180 - 120$

$y = 60$. 01

3 (b



Let $b_1 = b$

$b_2 = b + 4$ ... 01

$A = \frac{1}{2}(b_1 + b_2)h$ 01

$236 = \frac{1}{2}(b + b+4) \times 8^4$ 01

$236 = (2b + 4)4$

$236 \div 4 = 2b + 4$

$59 = 2b + 4$

$59 - 4 = 2b$

$\frac{55}{2} = \frac{2b}{2}$

$27.5 = b$

$b_1 = 27.5$ cm 01

$b_2 = 27.5 + 4 = 31.5$ cm. 01

4 (a) Let number of Girls be G, and Boys be B

$\begin{cases} G + B = 36 \\ 2G - B = 12 \end{cases}$ 01

by elimination

$+\begin{cases} G + B = 36 \\ 2G - B = 12 \end{cases}$

$\frac{3G}{3} = \frac{48}{3}$

$G = 16$ ... 02

From $G + B = 36$

$B = 36 - 16 = 20$ ... 01

Number of Girls = 16 and Boys = 20.

4 (b) $2y^2+5y+3=0$

$\frac{2y^2}{2}+\frac{5y}{2}+\frac{3}{2}=\frac{0}{2}$

$y^2+\frac{5y}{2}=-3/2$ ... 01

$y^2+\frac{5y}{2}+(\frac{5}{4})^2=-\frac{3}{2}+(\frac{5}{4})^2$ ... 00½

$(y+5/4)^2=-\frac{3}{2}+\frac{25}{16}$ ... 00½

$-11-=\frac{-24+25}{16}$

$(y+5/4)^2=\frac{1}{16}$

$\sqrt{(y+5/4)^2}=\sqrt{\frac{1}{16}}$

$y+\frac{5}{4}=\pm\frac{1}{4}$

$y=-\frac{5}{4}\pm\frac{1}{4}$ ... 01

Either $y=\frac{-5-1}{4}$ or $y=\frac{-5+1}{4}$

$y=\frac{-6}{4}$ or $y=\frac{-4}{4}$

$y=-\frac{3}{2}$ or $y=-1$ ... 02

5 (a) A : B : C

(A : B = 3 : 5)x2 ... 01

(A : C = 6 : 7)x1 ... 01

A : B = 6 : 10

A : C = 6 : 7

∴ A : B : C = 6 : 10 : 7 ... 03

5 (b) 180,000/= → 100%
x → 140%

$x = \frac{180,000 \times 140}{100}$

$= 252,000/=$

The selling price = 252,000/=.

6 (a) $|2x+1| > 3$

$\pm(2x+1) > 3$

$2x+1 > 3$ and $-(2x+1) > 3$

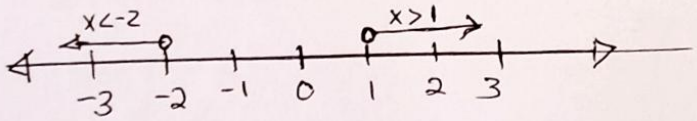
$2x > 3-1$ $\quad -2x-1 > 3$

$\frac{2x}{2} > \frac{2}{2}$ $\quad -2x > 3+1$

$x > 1$ $\quad -2x > 4$

$\frac{-2x}{-2} < \frac{4}{-2}$

$x < -2$



(b) (-2, 1) (4, -1)

$m = \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{-1-1}{4--2}$

$= \frac{-2}{6} = \frac{-1}{3}$

$\frac{y-1}{x--2} = -\frac{1}{3}$

$y-1 = -\frac{1}{3}(x+2)$

$y = -\frac{1}{3}x - \frac{2}{3} + 1$

$y = \frac{-x}{3} + \frac{-2+3}{3}$

$y = \frac{-x}{3} + \frac{1}{3}$

when $x = 0$

y-int $= \frac{1}{3}$

when $y = 0$

$0 = -\frac{x}{3} + \frac{1}{3}$

$3 \times \frac{x}{3} = \frac{1}{3} \times 3$

$x = 1$

x int = 1

7 (a) $\frac{5\sqrt{3}}{3\sqrt{2}+\sqrt{3}}$

$\frac{5\sqrt{3}(3\sqrt{2}-\sqrt{3})}{(3\sqrt{2}+\sqrt{3})(3\sqrt{2}-\sqrt{3})}$ ---- 01

$\frac{15\sqrt{6}-5(3)}{(3\sqrt{2})^2-(\sqrt{3})^2}$

$\frac{15\sqrt{6}-15}{18-3}$

$\frac{15\sqrt{6}-15}{15}$ ---- 01

$\frac{\cancel{15}(\sqrt{6}-1)}{\cancel{15}_1}$

$\sqrt{6}-1$ ---- 01

(b) $T = 2\pi\sqrt{\frac{x}{L}}$

$\frac{T}{2\pi} = \frac{\cancel{2\pi}}{\cancel{2\pi}}\sqrt{\frac{x}{L}}$ ---- 01

$\left(\frac{T}{2\pi}\right)^2 = \left(\sqrt{\frac{x}{L}}\right)^2$ ---- 01

$\frac{T^2}{4\pi^2} = \frac{x}{L}$ ---- 01

$L \times \frac{T^2}{4\pi^2} = \frac{x}{\cancel{L}} \times \cancel{L}$

$x = \frac{LT^2}{4\pi^2}$ ---- 01

or

$x = L\left(\frac{T}{2\pi}\right)^2$

7 (c) $(2^x)(5^y) = 100$

$2^x 5^y = 4 \times 25$

$2^x 5^y = 2^2 \times 5^2$ 01

then

$2^x = 2^2$

$\underline{x = 2}$ 01

and

$5^y = 5^2$

$\underline{y = 2}$ 01

8 (a) $\log_y 81 - \log_2 32 = -1$

$\log_y 81 - \log_2 2^5 = -1$

$\log_y 81 - 5\log_2 2 = -1$ 01

$\log_y 81 - 5 = -1$

$\log_y 81 = -1 + 5$

$\log_y 81 = 4$ 01

$y^4 = 81$

$y^4 = 3^4$

$y = 3$ 01

(b) $\log \sqrt[3]{x}$, $\log x = \bar{8}.0524$.

$\log \sqrt[3]{x} = \log x^{\frac{1}{3}}$

$\log \sqrt[3]{x} = \frac{1}{3}\log x$

$= \frac{1}{3}(\bar{8}.0524)$ 01

$= \frac{1}{3}(\bar{8} + \bar{1} + 1.0524)$

$= \frac{\bar{9}}{3} + \frac{1.0524}{3}$

$= \bar{3} + 0.3508$

$\log \sqrt[3]{x} = \bar{3}.3508$ 01

8 (c) $\log\left(\frac{764 \times 36.56}{\sqrt{3845}}\right)$, $\log 7.64 = 0.8831$

$\log 3.656 = 0.5630$

$\log 3.845 = 0.5849$

$\log 764 + \log 36.56 - \log \sqrt{3845}$ C1

$\log(7.64 \times 10^2) + \log(3.656 \times 10^1) - \log(3.843 \times 10^3)^{\frac{1}{2}}$ C1

$\log 7.64 + \log 10^2 + \log 3.656 + \log 10^1 - \frac{1}{2}(\log 3.843 + \log 10^3)$

$2 + 0.8831 + 0.5630 + 1 - \frac{1}{2}(0.5849 + 3)$ C1

$2.8831 + 1.5630 - \frac{1}{2}(3.5849)$

$4.4461 - 1.7925$

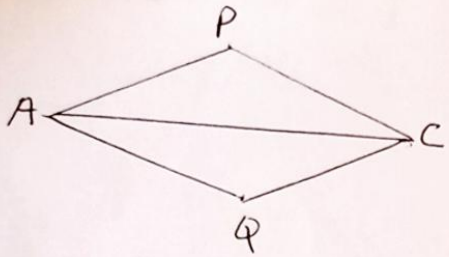
2.6536 C1

Or

No	S.F	Log	
764	7.64×10^2	2.8831	C1
36.56	3.656×10^1	1.5630	C½
N		4.4461	C1
$\sqrt{3846}$	3.846×10^3	$\frac{3.5849}{2}$	C½
D		1.7925	C1
N/D		4.4461 − 1.7925	
		2.6536	C1

$\therefore \log\left(\frac{764 \times 36.56}{\sqrt{3845}}\right) = 2.6536$

9 (a)



AC bisects $P\hat{A}Q$ and $P\hat{C}Q$ -- given

$P\hat{A}C = C\hat{A}Q$ ($\overline{AC}$ bisect $P\hat{A}Q$) --- A ---- 01

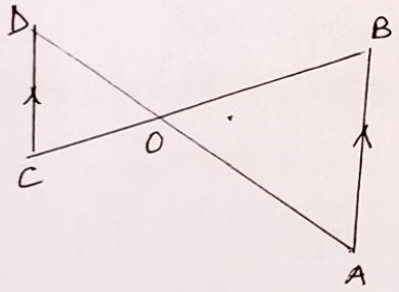
$P\hat{C}A = A\hat{C}Q$ ($\overline{AC}$ bisect $P\hat{C}Q$) --- A ---- 01

$\overline{AC}$ = Common ---- S ---- 01

$\therefore \triangle APC \cong \triangle AQC$ (AAS postulate) --- 01

hence $\overline{AP} = \overline{AQ}$ (definition of congruence of triangles) -- 01

(b)



$\overline{CD} \parallel \overline{AB}$ --- given

$C\hat{D}O = B\hat{A}O$ (alternate interior angles $CD \parallel AB$) -- A

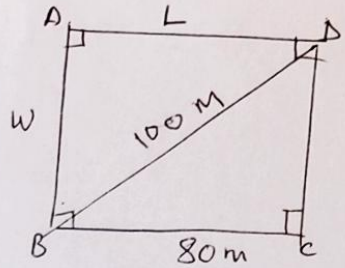
$D\hat{C}O = O\hat{B}A$ (-11- -11-) -- A ---- 01

$D\hat{O}C = A\hat{O}B$ (opposite angles) -- A ---- 01

$\therefore \triangle DOC \sim \triangle BOA$ (equiangular triangles) ---- 02
or AA-similarity theorem,

10 L = 80m

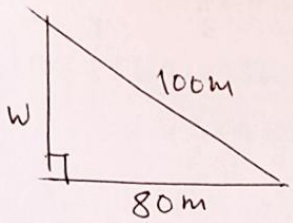
diagonal = 100 M



01

10

(a)



by pythagorus theorem

$a^2 + b^2 = c^2$ ---------- 01

$w^2 + 80^2 = 100^2$ ---------- 01

$w^2 = 100^2 - 80^2$

$w^2 = (100+80)(100-80)$

$= 180 \times 20$

$w^2 = 3600$ ---------- 01

$w = 60m$ ---------- 01

(b) Length of the wire = Perimeter of the rectangular field

$P = 2(L+w)$ ---------- 01

$= 2(80+60)$ ---------- 02

$= 2 \times 140$

$= 280 M$

∴ Length of the wire $= 280 M$ ---------- 02